Literature Status for Problem 14 of arXiv:1207.2958

FA/Banach run packet

June 14, 2026

Status

Result type: literature already answered.

Source problem: Godefroy–Lancien–Zizler, *The non-linear geometry of Banach spaces after Nigel Kalton*, arXiv:1207.2958, Problem 14.

Supporting answer: Baudier–Lancien–Schlumprecht, *The coarse geometry of Tsirelson's space and applications*, arXiv:1705.06797.

Open Problem Evidence

The source survey asks the following question in Problem 14.

> Banach spaces.
>
> For the record, we recall the
>
> **Problem 14.** Does ℓ_2 coarsely embed into every infinite dimensional Banach space?
>
> 7. Lipschitz-free spaces and their applications

Transcription:

> Does ℓ_2 coarsely embed into every infinite dimensional Banach space?

The word "infinite" is part of the source statement. This packet is therefore not based on a finite-dimensional reading.

Supporting Literature Evidence

The later paper explicitly identifies the same question as Problem 14 in the survey and restates it as the Main Problem.

> sequently, whether the separable Hilbert space is the Banach space into which it is the hardest to embed, became a very natural and intriguing question. More precisely, the following problem was raised (Problem 14 in [15], Problem 11.17 in [34]).
>
> **Main Problem.** *Does ℓ_2 coarsely embed into every infinite dimensional Banach space?*

The same paper then defines a Banach space to be coarsely minimal when it coarsely embeds into every infinite-dimensional Banach space and records the negative conclusion.

We will say that a Banach space is *coarsely minimal* if it coarsely embeds into every infinite dimensional Banach space. The Main Problem asks whether ℓ_2 is coarsely minimal. In fact, Theorem A provides a much stronger negative solution to the Main Problem. Indeed, a coarsely minimal Banach space embeds into ℓ_2 and it must have non-trivial cotype using Mendel and Naor metric cotype notion [24], but it also embeds into T^* and by Theorem A it must have trivial cotype, a contradiction.

Corollary C. *There is no coarsely minimal infinite dimensional Banach space.*

Idea of the Proof

The identification is direct rather than inferential. The source asks whether ℓ_2 is universal for coarse embeddability into all infinite-dimensional Banach spaces. Baudier–Lancien–Schlumprecht cite that exact problem, call such universal spaces "coarsely minimal", and prove that no infinite-dimensional Banach space is coarsely minimal. Their argument uses Tsirelson's original space T^*: a concentration inequality for infinite Hamming graphs in T^* obstructs coarse embeddings of spaces whose spreading-model or metric cotype structure is incompatible with T^*. In particular, ℓ_2 fails to coarsely embed into T^*.

Formal Statement

Theorem. *Problem 14 of Godefroy–Lancien–Zizler, arXiv:1207.2958, has a negative answer in the later literature. The Hilbert space ℓ_2 does not coarsely embed into every infinite-dimensional Banach space.*

Proof. Baudier–Lancien–Schlumprecht state that the question whether ℓ_2 coarsely embeds into every infinite-dimensional Banach space was raised as Problem 14 in [1]. This is the same question as the source Problem 14 displayed above.

Their Corollary B says that, for every $p \in [1, \infty)$, the space ℓ_p does not coarsely embed into Tsirelson's original space T^*. Taking $p = 2$, ℓ_2 does not coarsely embed into T^*. Since T^* is an infinite-dimensional Banach space, this gives a negative answer to the source question. Equivalently, their Corollary C states the stronger conclusion that there is no coarsely minimal infinite-dimensional Banach space. □

Verification and Limitations

The source PDF is stored as `source_paper.pdf`. The supporting answer PDF is stored as

`supporting_paper_1705.06797.pdf`.

The crops above come from page 33 of the source PDF and pages 2 and 4 of the supporting PDF.

This packet does not answer the later final question in [2] asking whether ℓ_2 coarsely embeds into every super-reflexive Banach space. It also does not use the run's human-only archive for the previously rejected finite-dimensional literal reading of that later question.

Search bounds: before packaging, the run's registry, solutions index, attempts index, and proof-gaps index were searched for arXiv:1207.2958, Problem 14, coarsely minimal, and the ℓ_2/every infinite-dimensional Banach-space wording. A web/arXiv phrase search for the exact question found arXiv:1705.06797, and its local TeX source was checked for the explicit Problem 14 citation and negative conclusion.

References

References

[1] G. Godefroy, G. Lancien, and V. Zizler, *The non-linear geometry of Banach spaces after Nigel Kalton*, Rocky Mountain J. Math. 44 (2014), no. 5, 1529–1583. arXiv:1207.2958. https://arxiv.org/abs/1207.2958

[2] F. Baudier, G. Lancien, and Th. Schlumprecht, *The coarse geometry of Tsirelson's space and applications*, J. Amer. Math. Soc. 31 (2018), no. 3, 699–717. arXiv:1705.06797. https://arxiv.org/abs/1705.06797